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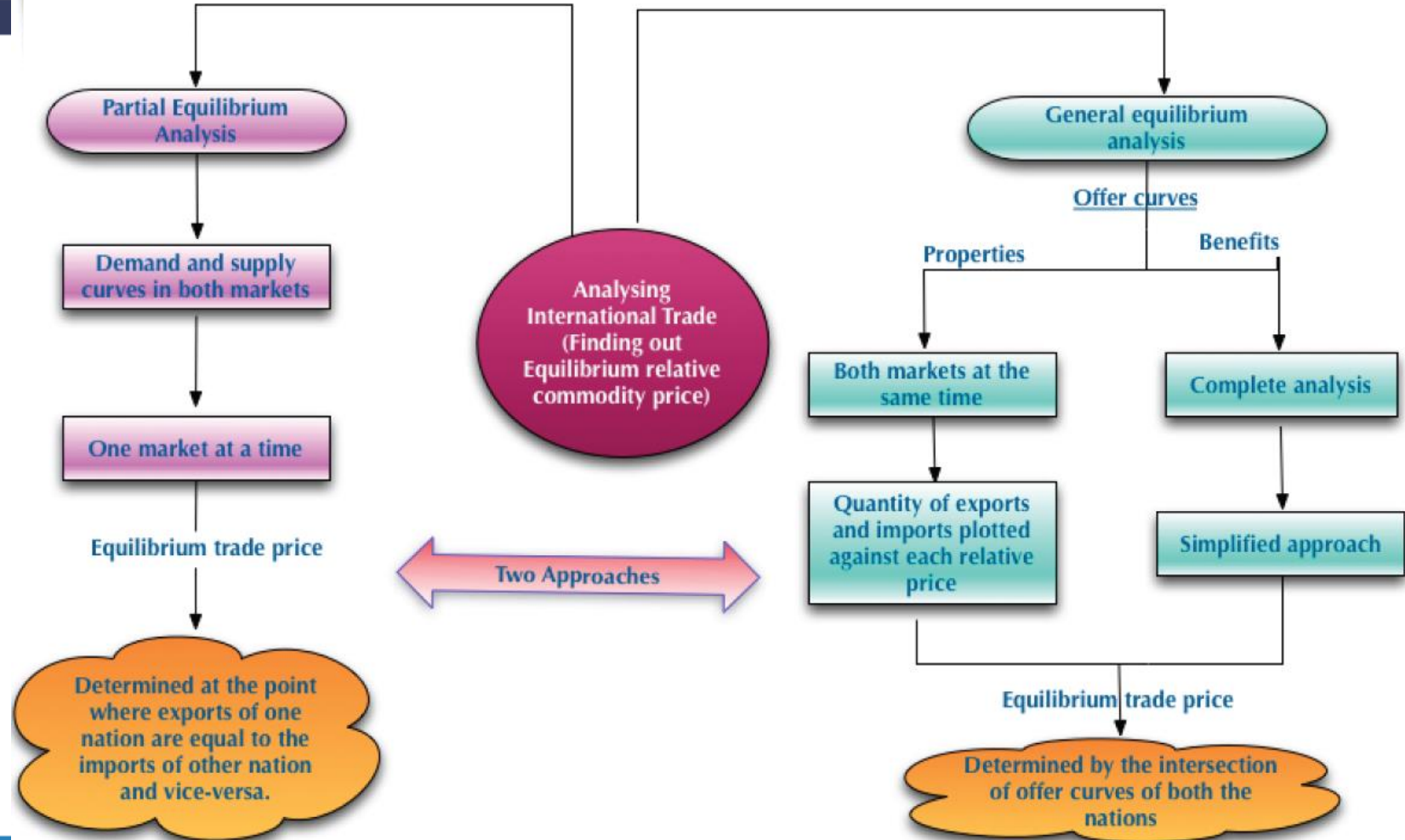
NPTEL ONLINE
CERTIFICATION COURSE

Partial Equilibrium Analysis of Trade

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Introduction

- Relative commodity price differences between two nations in isolation reflect comparative advantage, and forms basis for mutually beneficial trade.
- Can use partial and general equilibrium analysis to determine equilibrium-relative commodity price at which trade will take place.
- partial equilibrium implies that the analysis only considers the effects of a given policy action in the market(s) that are directly affected. That is the analysis does not account for the economic interactions between the various markets in a given economy. In a general equilibrium setup all markets are simultaneously modelled and interact with each other.
- **SMART**, the market access simulation package included in WITS, is a partial equilibrium modelling tool.
- **GTAP** for general equilibrium



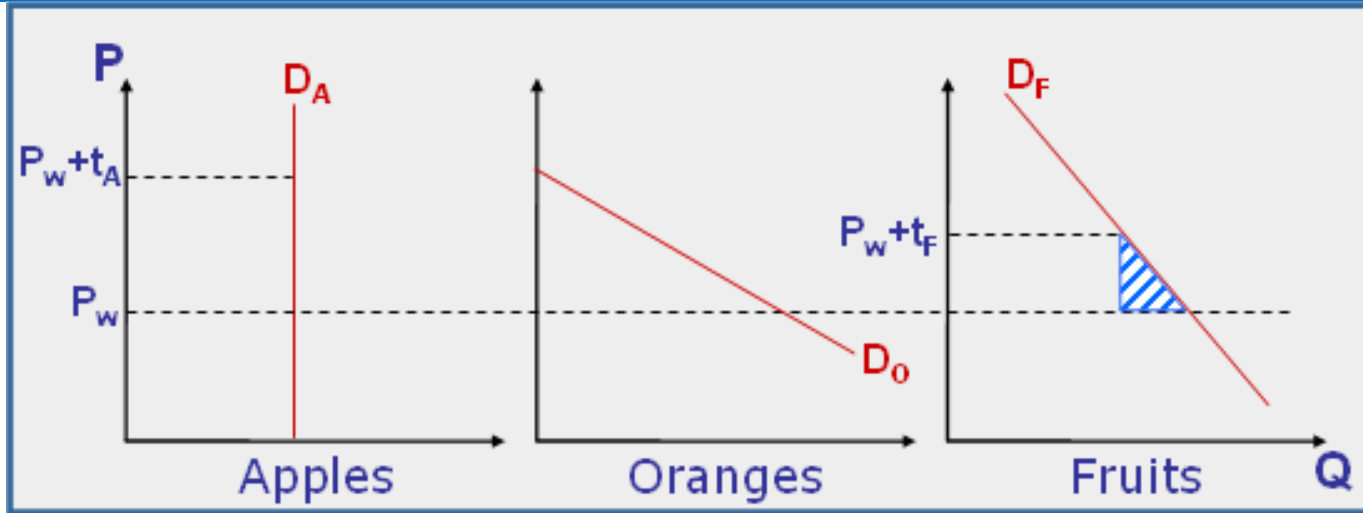
The advantages of partial equilibrium modelling

- minimal data requirement
- the only required data for the trade flows, the trade policy (tariff), and a couple of behavioural parameters (elasticities)
- WITS datasets which contain all of those
- analysis at a fairly disaggregated (or detailed) level
 - For example, it allows the study of the effects of the liberalization of “brown rice”
- This also resolves a number of “aggregation biases”



- By the same logic, the partial equilibrium may allow the analysis of the likely impact of trade agreements like the Doha round more accurately, as the negotiations are conducted at a very disaggregated level.
- In general, by virtue of their simplicity, partial equilibrium models tend to be more transparent and easy to implement. Modelling is straightforward and results can be easily explained.

Illustrating the aggregation bias



The graphics above illustrates an aggregation bias by considering 2 products (Apples and Oranges) falling into the same product category (Fruits). Apples face a tariff t_A but demand for apples (D_A) is perfectly inelastic. Consequently, t_A does not entail welfare cost on the Apples market since it does not affect the level of imports. On the Oranges market, demand is somehow elastic but imports face no tariff. Again, there is no welfare cost. Therefore, the protection on the Fruits market is clearly not associated with welfare cost when analyzing at its component level, while analysis implemented at the Fruits (aggregated) level would conclude that there is welfare cost the blue stripe triangle because of the aggregation bias.

The disadvantages of partial equilibrium modelling

- The partial equilibrium approach also has a number of disadvantages that have to be kept in mind while conducting any analysis. Since it is only a “partial” model of the economy, the analysis is only done on a pre-determined number of economic variables. This makes it very sensitive to a few (badly estimated) behavioural elasticities.
- Due to their simplicity also, partial equilibrium models may miss important interactions and feedbacks between various markets. In particular, the partial equilibrium approach tends to neglect the important inter-sectoral input/output (or upstream/downstream) linkages that are the basis of general equilibrium analyses. It also misses the existing constraints that apply to the various factors of production (e.g., labor, capital, land...) and their movement across sectors.



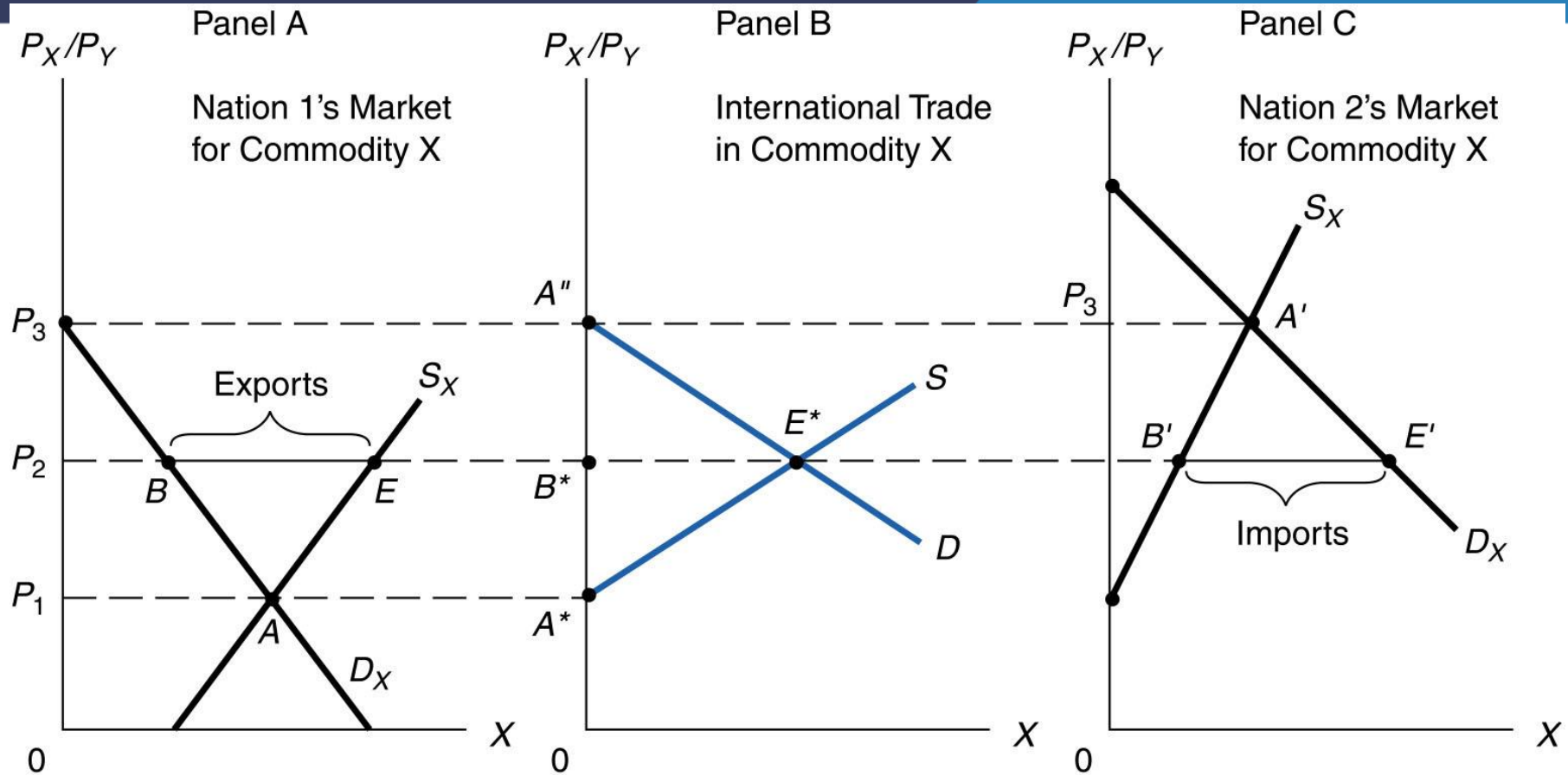


FIGURE The Equilibrium-Relative Commodity Price with Trade with Partial Equilibrium Analysis.

The Equilibrium-Relative Commodity Price with Trade-Partial Equilibrium Analysis

- At a relative price greater than P_1 , Nation 1's excess supply of X (Panel A) gives rise to Nation 1's international supply curve of X (S in Panel B).
- At a relative price lower than P_3 , Nation 2's excess demand for X (Panel C) gives rise to Nation 2's demand for imports of X (D in Panel B).
- Only at P_2 (Panel B) does quantity of imports demanded equal quantity of exports supplied.
- Thus P_2 is equilibrium-relative commodity price with trade.



Offer Curves

- **Offer curves** (sometimes called **reciprocal demand curves**) introduced to international economics by *Marshall* and *Edgeworth*.
- Shows how much of its import commodity a nation demands for it to be willing to supply various amounts of its export commodity.
- Can be derived from production possibilities frontier, indifference map and various hypothetical relative commodity prices at which trade could take place.

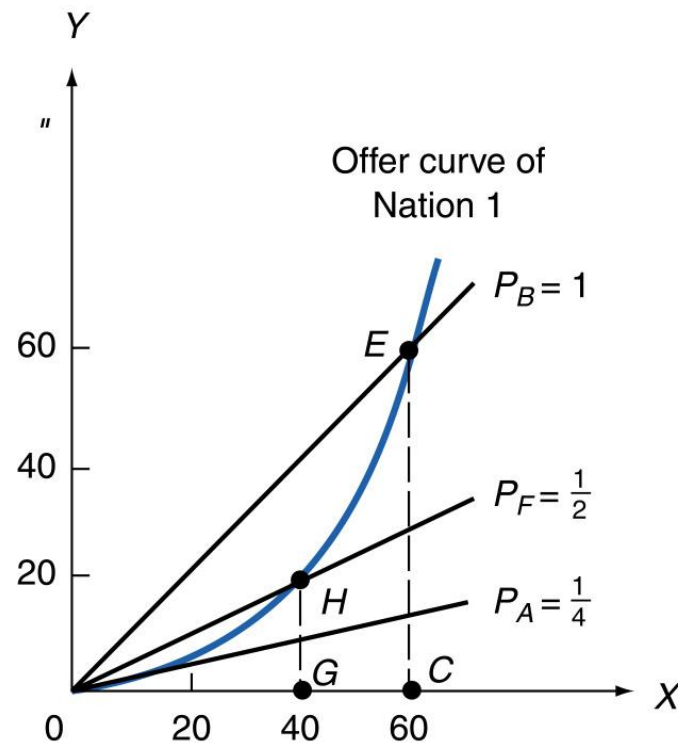
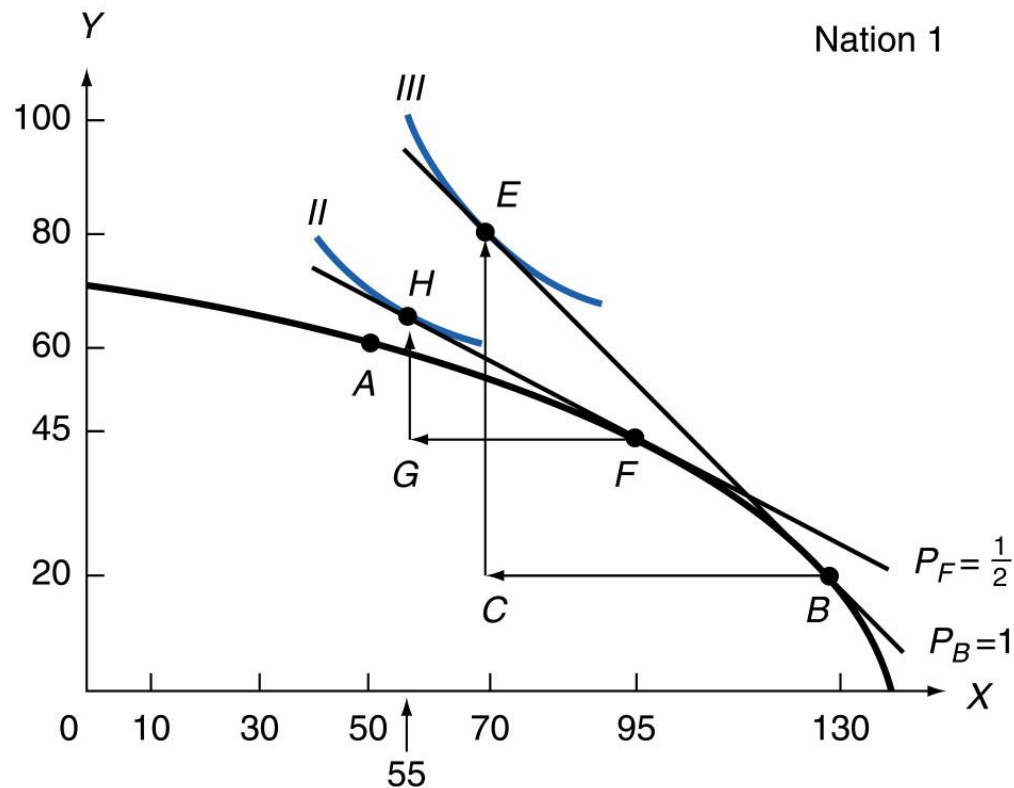


FIGURE Derivation of the Offer Curve of Nation 1.

Convex offer curve because: (1) Nation 1 incurs increasing opportunity costs in production of X for exports

(2) the more of Y and less of X that Nation 1 consumes with trade

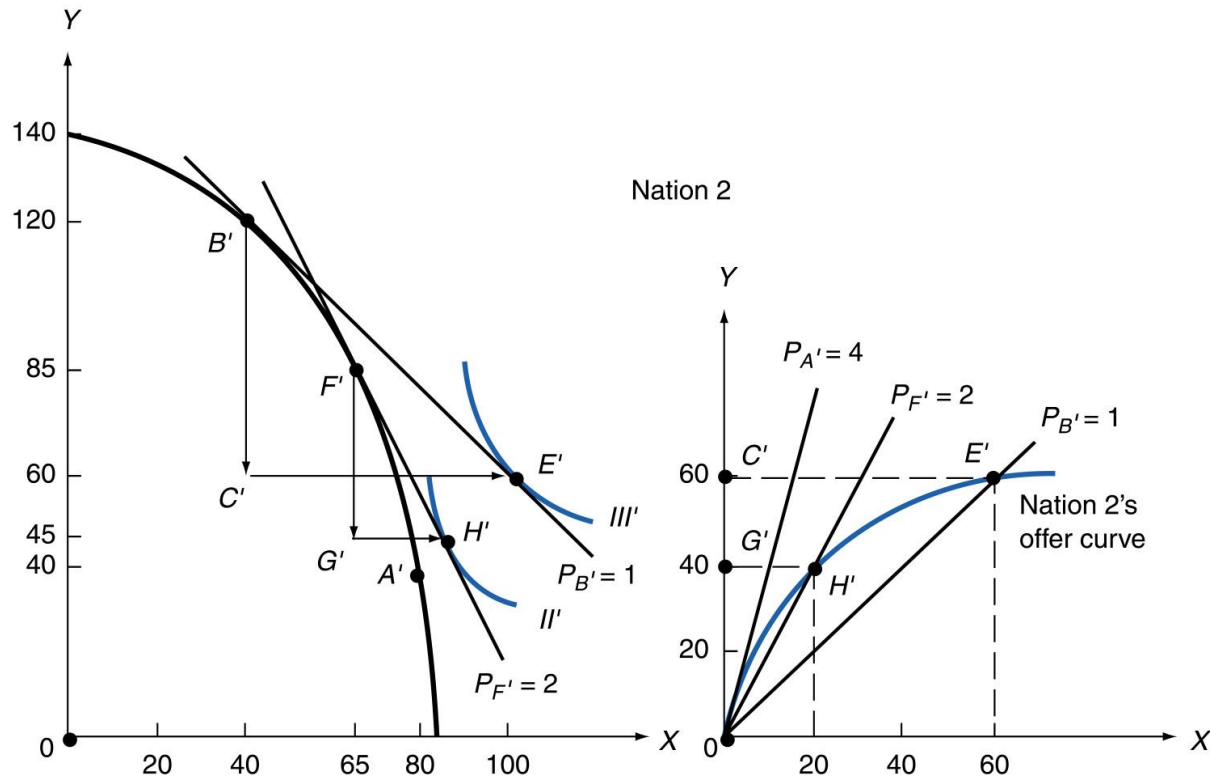


FIGURE Derivation of the Offer Curve of Nation 2.

The Equilibrium-Relative Commodity Price with Trade-General Equilibrium Analysis

- **Equilibrium-relative commodity price** with trade found at intersection of offer curves for two nations.
- Only at this equilibrium price will trade be balanced.
- At any other relative commodity price, quantities of imports do not equal quantities of exports, placing pressure on relative commodity price to move toward equilibrium.



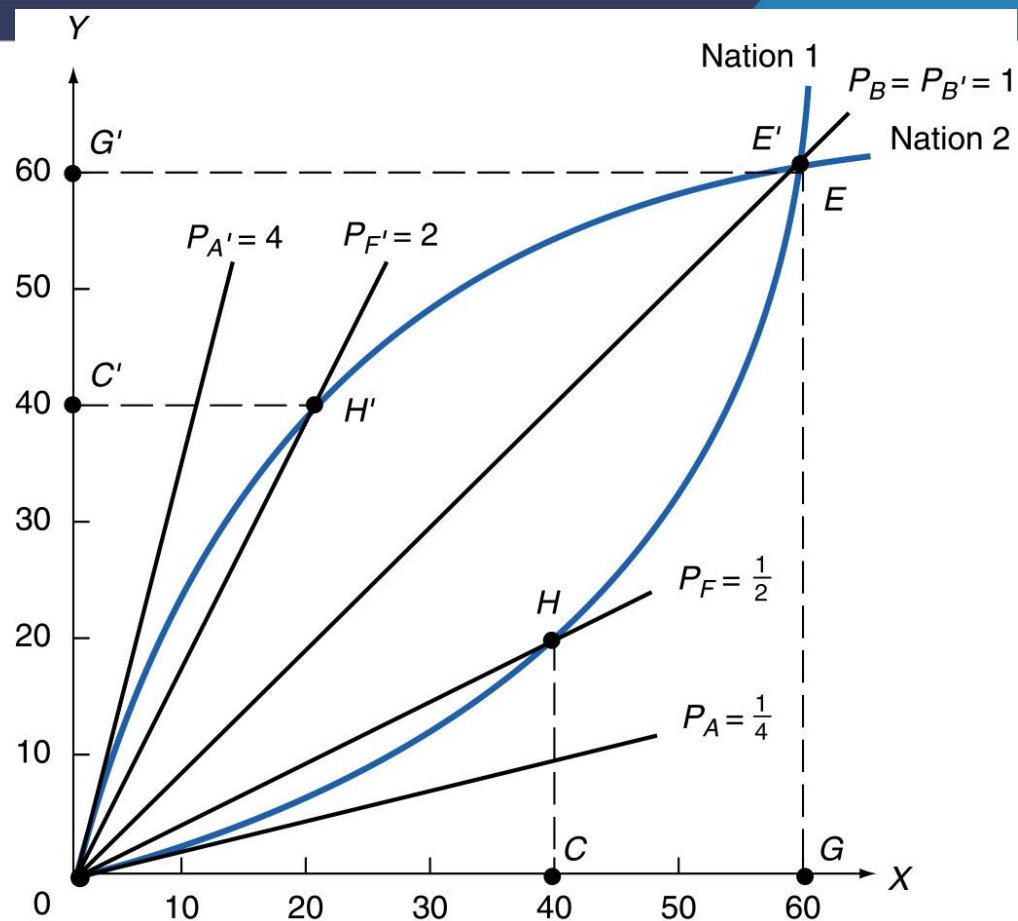


FIGURE Equilibrium-Relative Commodity Price with Trade.

Relationship between General and Partial Equilibrium Analyses

- Both partial equilibrium and general equilibrium analysis use production frontiers and indifference maps to find equilibrium trade price.
- Only general equilibrium analysis considers all markets together, not just the market for commodity X.
- Changes in the market for X affect other markets, which possibly impact the market for X.
- General equilibrium analysis is therefore required for more complete analysis.



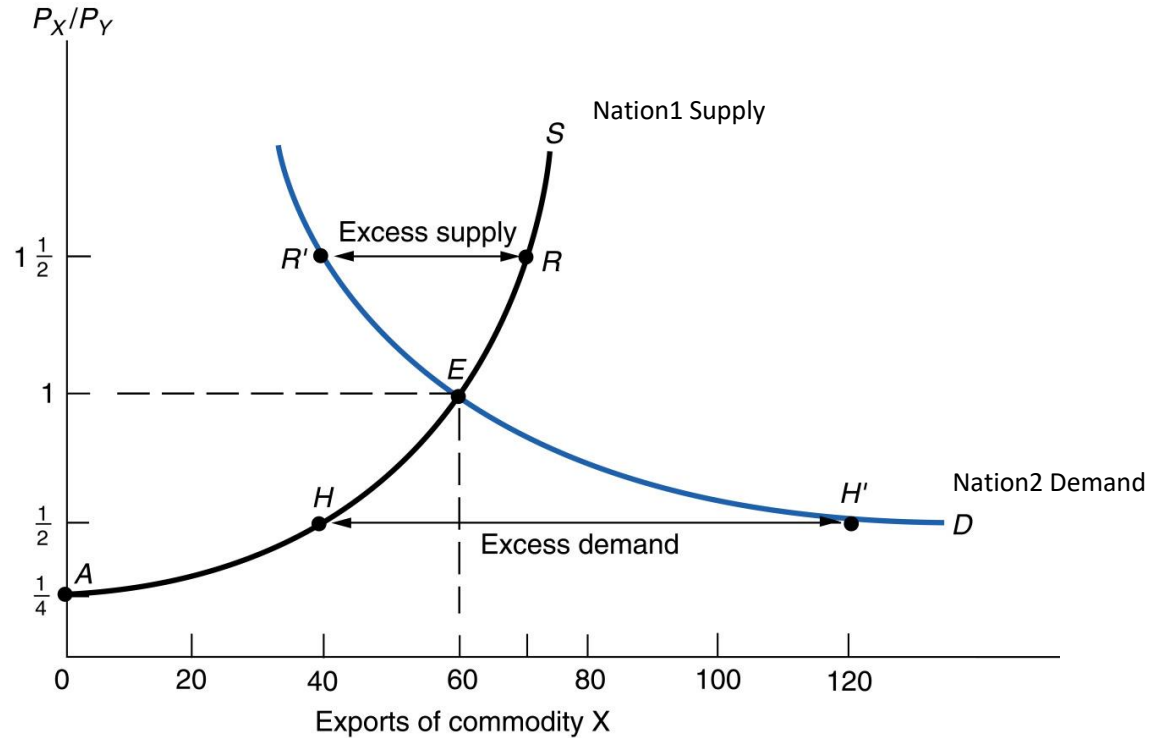


FIGURE Equilibrium-Relative Commodity Price with Partial Equilibrium Analysis.

(partial since only commodity X has been considered)

The Terms of Trade

- **Terms of trade** = the ratio of the price of a nation's export commodity to the price of its import commodity.
 - In a two-nation world, the terms of trade of Nation 1 are equal to the reciprocal of the terms of trade of Nation 2.
 - In a world of many traded goods, the terms of trade is the ratio of the export price index to the import price index, also called **commodity** or **net barter terms of trade**.
 - If Nation 1 exports X and imports Y, its terms of trade are given by P_X/P_Y , where P = price index.



*Thank
you*

